



Differential Privacy in Movie Revenue Dataset

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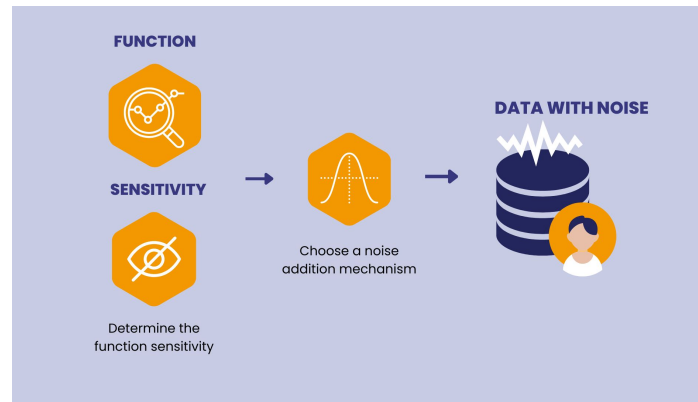
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What is Differential Privacy?

- The usage of statistics in order to balance the level of privacy given to users in a database and the relevancy of the data to be used by researchers
- 3 important features
 - Initial accuracy of the data collected
 - What noise to add to a model
 - Amount of queries available (aka how much information asked of the database)
- 2 variations
 - Local: noise added to individual points
 - Global: noise is added to all inputs of a database at once through a data curator



Noise Implementation Methods



1. Laplace mechanism
2. Gaussian mechanism
3. Exponential mechanism

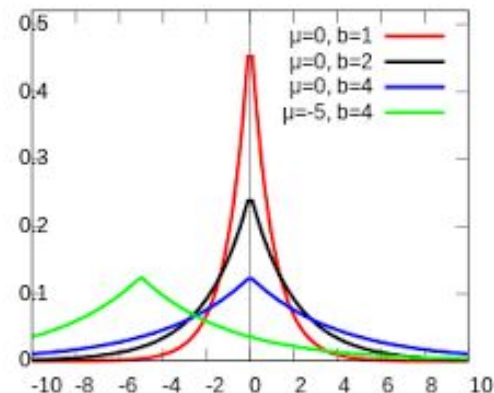
How to Implement Differential Privacy: Laplace

Laplace ($\epsilon, 0$)

- Include noise from a Laplace distribution into database query results

$$f(x|\mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right) \quad b = \frac{\Delta f}{\epsilon}$$

- μ : the mean
 - b : the scale parameter
 - Δf : L1 sensitivity of the function f
 - ϵ : privacy parameter
- Adds positively associated noise to L1 sensitivity
 - Include noise negatively related to privacy parameter
 - Smaller ϵ value, increase the scale, more noise $\rightarrow$ stronger privacy
 - Disadvantages
 - Not suited to high-sensitivity queries
 - Only apply to numerical values



How to Implement Differential Privacy: Gaussian Mechanism

Gaussian Mechanism(ϵ, δ)

- Adding noise from a normal distribution with 2 parameters: mean and sd

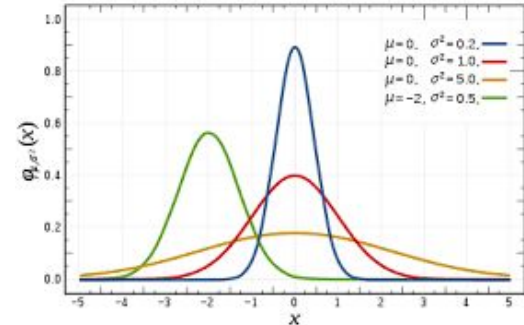
$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- μ : the mean
- σ : the standard deviation

$$\sigma = \frac{\Delta f \sqrt{2 \ln(1.25/\delta)}}{\epsilon}$$

- Δf : sensitivity of the function f
- ϵ : privacy parameter
- δ : allows small probability of mechanism

- Difference with Laplace: Gaussian can use both L1 and L2 sensitivity instead of only L1
- Differences of Gaussian Mechanism
 - Add less noise due to L2's lowered sensitivity
 - Uses a different formula for L2 compared to L1
- Disadvantages
 - Only apply for numerical queries



How to Implement Differential Privacy: Exponential Mechanism

Exponential Mechanism ($\epsilon, 0$)

- Selects output/data with high scores or outcome based on utility function $u(D, r)$ to gain relevant data that ensures differentiated privacy
 - Works on both numerical and categorical queries
 - Chooses scores based on their best balance of high score and potential to protect privacy

$$\Pr[r \text{ is chosen}] \propto \exp\left(\frac{\epsilon \cdot u(D, r)}{2\Delta u}\right)$$

– Δu = sensitivity of utility function

– ϵ : privacy parameter

- Disadvantages
 - Difficult implementation
 - High sensitivity
 - Complex utility functions often too hard to define

Examples of Differential Privacy

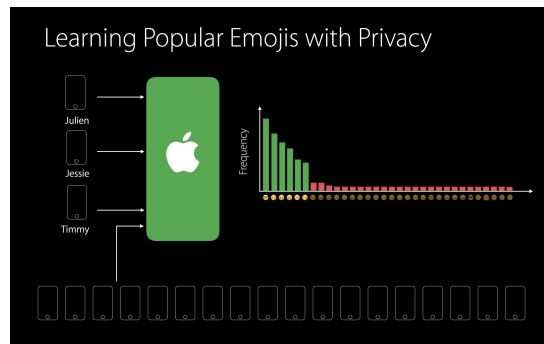
US Census

- Used to protect individuals' identities when determining state populations
- Swapping people between blocks thus keeping people that live in a block anonymous
- Led to places such as rural areas to have variances increase



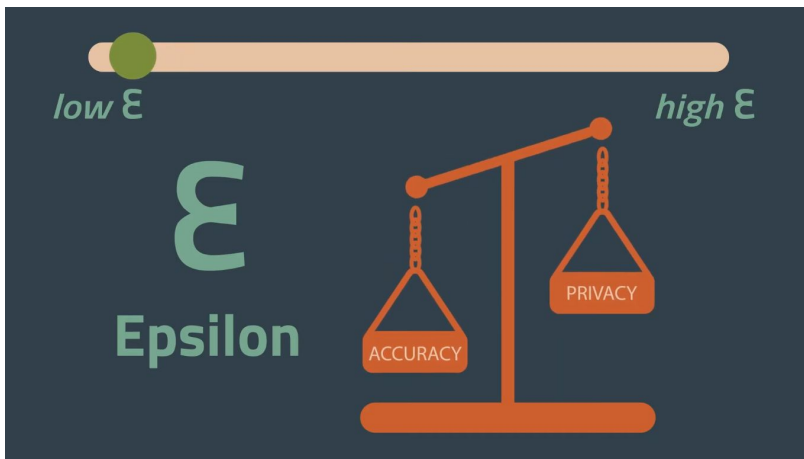
Apple

- Used to gather insight from users to improve their ui while keeping responses confidential
- Implemented local differential privacy through hashing, subsampling, and noise injection
- Data gathered from emoji suggestions, dictionary updates, spotlight search, and lookup hints



Motivation/Goals for Differential Privacy

- Understanding how differential privacy protects our sensitive information
- How to implement differential privacy into our own project
- Understand the positives and negatives of differential privacy
- How to optimize DP to maximize benefits for both the researchers and people in a database



Data Introduction



- The original dataset *movies* was downloaded from Kaggle.
- This project aims to deepen the understanding of how different factors may impact a movie's revenue.
- Hypotheses:
 - Different attributes of a feature movie significantly impacts its revenue.
 - A movie's budget and genre contribute more significantly than others.
 - The model will become less powerful as the amount of noise increases

Data Description



- **Response variable**
 - *revenue* (measured in thousands of dollars): the amount of money a movie raised by ticket sales, numerical
- **Predictor variables**
 - *genres*: genre of the movie, categorical
 - *original language*: the language in which the movie was first released, categorical
 - *popularity*: the amount of times a movie's page has been visited according to TMDB metric, numerical
 - *budget* (measured in thousand dollars): the total projected costs of making a movie, numerical
 - *runtime* (measured in minutes): duration of the movie, numerical
 - *vote average* (on a scale from 0-10): average of votes given by TMDB users, numerical
 - *vote count*: total number of TMDB votes a movie received, numerical

DP Implementation into Movies

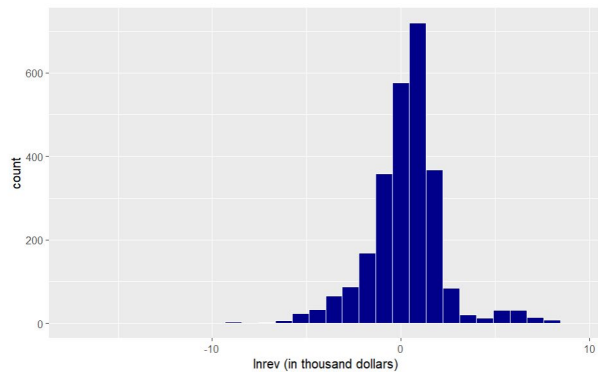
- Our original dataset had all NA variables removed
 - This means important information could have been discarded
- Step 1: replace NA values with values sourced from a normal distribution using the mean and standard deviation of the respective variables
 - Number of observations: 2916 – meaning we gained back 292 movies
- Step 2: Added noise to continuous variables through the jitter function with the amount of noise added equal to 1, 2, 5, and 10 to compare results

```
set.seed(1)
noisyy$budget <- jitter(noisyy$budget, amount = 10)
noisyy$vote_count <- jitter(noisyy$vote_count, amount = 10)
noisyy$popularity <- jitter(noisyy$popularity, amount = 10)
noisyy$revenue <- jitter(noisyy$revenue, amount = 10)
noisyy$vote_average <- jitter(noisyy$vote_average, amount = 10)
noisyy$runtime <- jitter(noisyy$runtime, amount = 10)
```

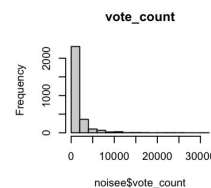
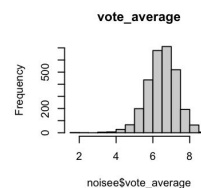
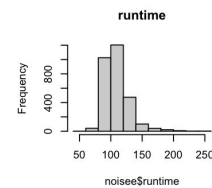
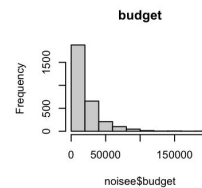
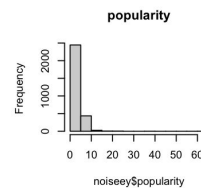
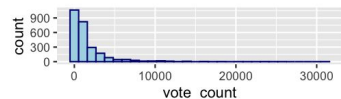
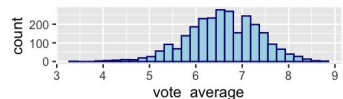
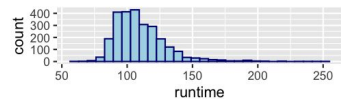
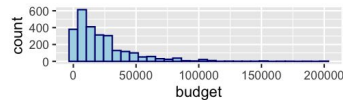
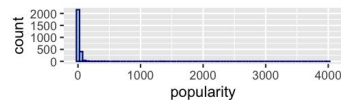
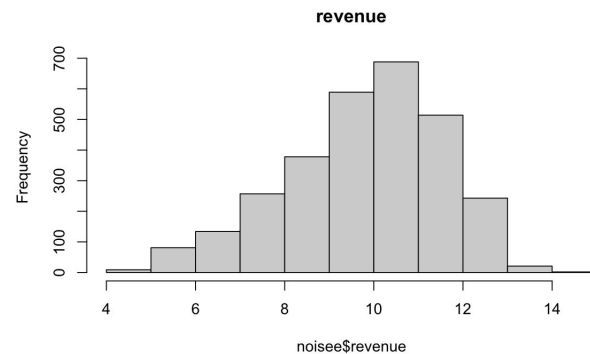
	X	genres	original_language	popularity	budget	revenue	runtime	vote_average	vote_count
	<int>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	Drama	en	54.31257	74993.70	21.910501	113.5181	-2.0851177	1121.761
2	2	Drama	en	35.41972	100004.05	5.880509	166.9863	1.9643995	1212.321
3	3	Comedy	en	15.68580	125001.47	12.345221	123.9641	5.6105149	2682.939
4	4	Comedy	en	21.32298	144993.36	22.285176	106.0238	1.0735675	7070.536
5	5	Comedy	en	27.78249	40008.88	1.045449	141.2186	16.1844882	3577.872
6	6	Drama	en	18.02642	100008.87	22.798646	172.4232	0.5245502	7604.864

Step I EDA

Original Data:

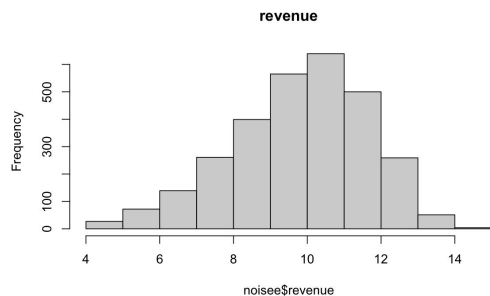


Data with noise in obs with NAs:

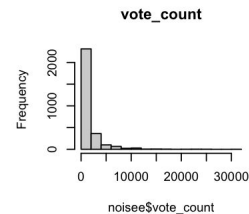
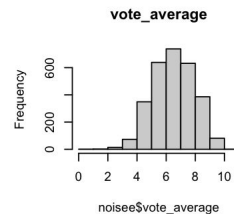
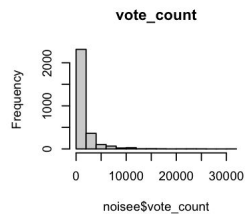
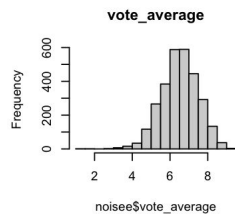
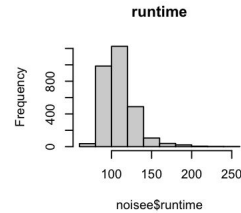
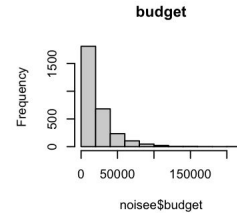
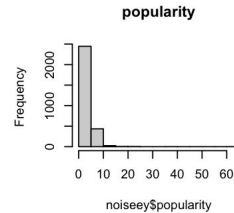
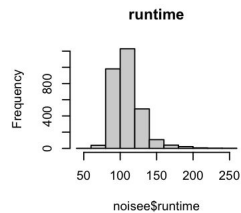
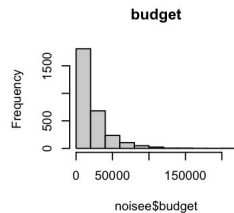
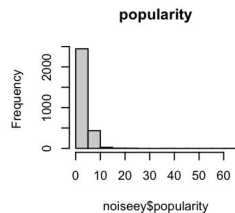
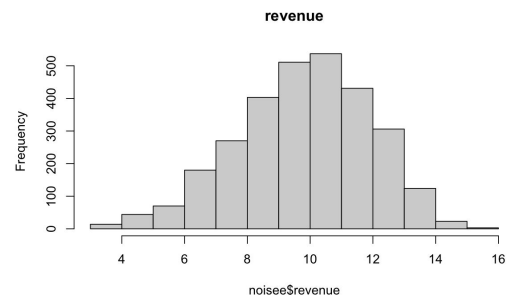


Step II EDA Pt. I

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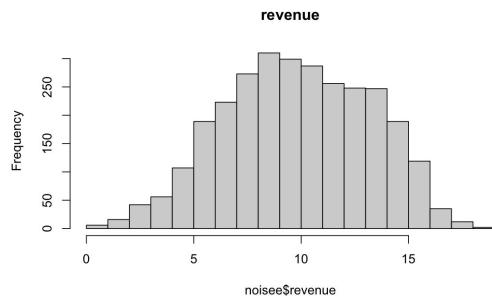


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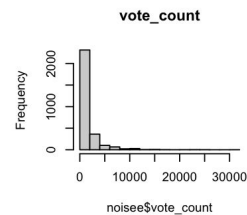
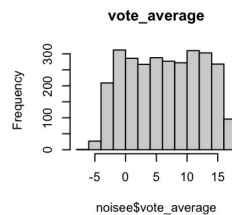
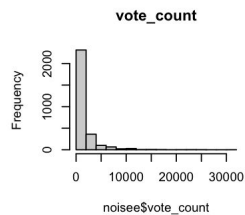
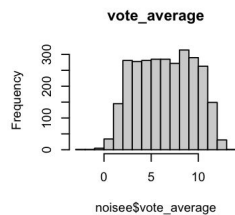
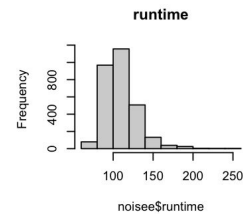
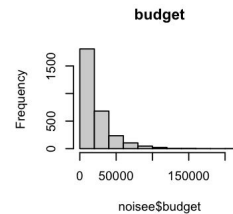
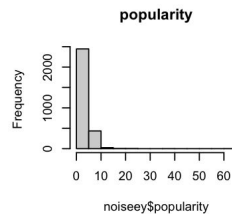
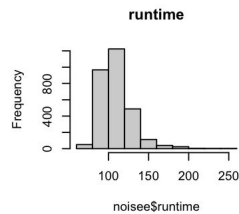
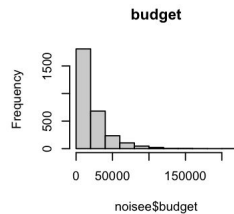
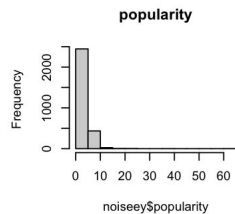
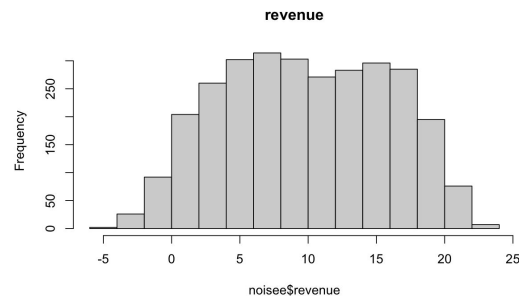


Step II EDA Pt. II

Amount = 5



Amount = 10



Correlation Table



Amount	Popularity	Budget	Runtime	Vote Avg	Vote Count
0	0.3924	0.5496	0.2267	0.1971	0.4795
5	0.1214	0.2880	0.0854	0.0456	0.2397
10	0.0300	0.1532	0.0566	0.0353	0.1661

Correlation Plots



Amount

Popularity

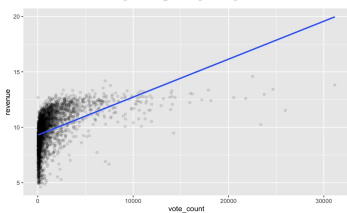
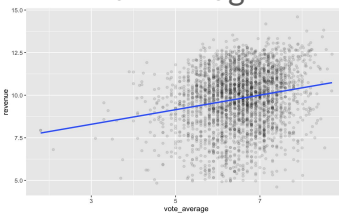
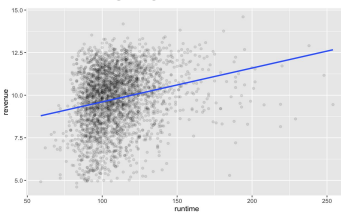
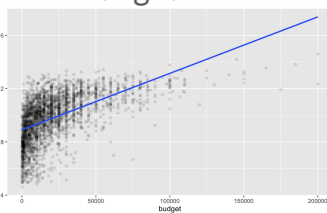
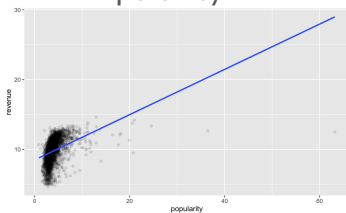
Budget

Runtime

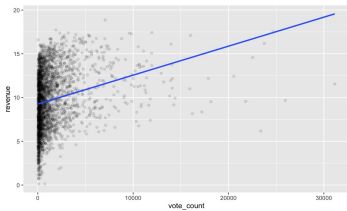
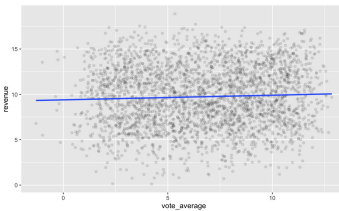
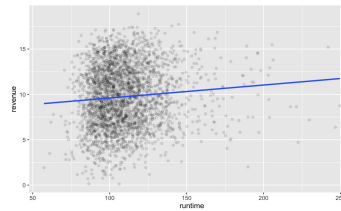
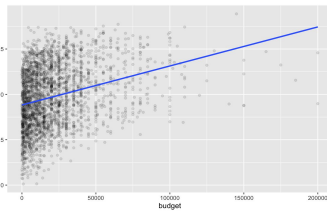
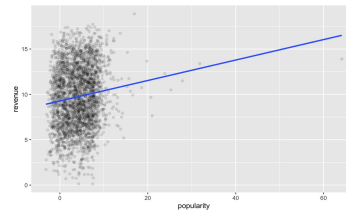
Vote Average

Vote Count

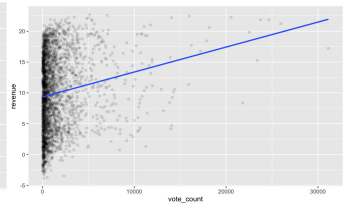
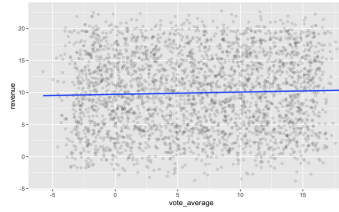
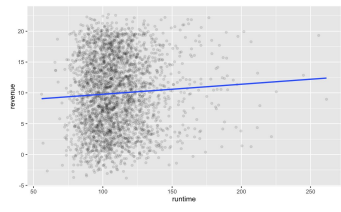
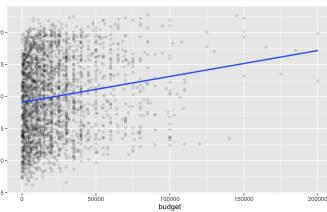
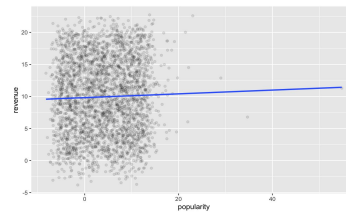
0



5



10



Multicollinearity

	GVIF	Df	$GVIF^{1/(2*Df)}$
budget	2.348206	1	1.532386
popularity	6.187330	1	2.487434
genre	2.787927	1	1.669709
vote_count	5.066213	1	2.250825
original_language	6.386995	2	1.589733
genre:vote_count	4.251180	1	2.061839

	GVIF	Df	$GVIF^{1/(2*Df)}$
budget	2.340698	1	1.529934
popularity	5.736339	1	2.395066
genre	2.786268	1	1.669212
vote_count	5.034497	1	2.243769
original_language	6.157974	2	1.575286
genre:vote_count	4.249377	1	2.061402

	GVIF	Df	$GVIF^{1/(2*Df)}$
budget	2.323245	1	1.524220
popularity	4.761268	1	2.182033
genre	2.785332	1	1.668932
vote_count	4.979488	1	2.231477
original_language	5.611982	2	1.539143
genre:vote_count	4.249725	1	2.061486

Data with noise in obs with NAs

Amount = 1

Amount = 2

	GVIF	Df	$GVIF^{1/(2*Df)}$
budget	2.268546	1	1.506169
popularity	2.474102	1	1.572928
genre	2.783636	1	1.668423
vote_count	4.784868	1	2.187434
original_language	4.421634	2	1.450092
genre:vote_count	4.235082	1	2.057932

Amount = 5

	GVIF	Df	$GVIF^{1/(2*Df)}$
budget	2.248301	1	1.499434
popularity	1.466675	1	1.211063
genre	2.783626	1	1.668420
vote_count	4.710898	1	2.170460
original_language	3.880945	2	1.403571
genre:vote_count	4.235723	1	2.058087

Amount = 10

All models have low VIF values, meaning that there is no multicollinearity issues. However, the numerical variables **VIF slightly decreases as noise increases**. We assume this is because correlation between variables might decrease due to noise.

Final Model: Original vs Noisy Data

```
## Call:
## lm(formula = lnrev ~ 0 + budget + popsqrt * genre + vc + o_lang)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.7908  -0.6587   0.4173   1.2191   8.9885
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## budget          2.637e-05  1.936e-06  13.620  < 2e-16 ***
## popsqrt         8.531e-01  3.021e-02  28.237  < 2e-16 ***
## genre           3.320e+00  1.622e-01  20.468  < 2e-16 ***
## vc              1.363e-04  1.825e-05   7.468  1.1e-13 ***
## o_lang           5.516e+00  1.273e-01  43.335  < 2e-16 ***
## popsqrt:genre -7.775e-01  3.581e-02 -21.715  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.07 on 2618 degrees of freedom
## Multiple R-squared:  0.9579, Adjusted R-squared:  0.9578
## F-statistic: 9929 on 6 and 2618 DF, p-value: < 2.2e-16
```

Original dataset

Coefficients:					Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)		Estimate	Std. Error	t value	Pr(> t)
budget	2.979e-05	1.331e-06	22.380	< 2e-16 ***	budget	3.043e-05	1.628e-06	18.692	< 2e-16 ***
popularity	9.383e-02	1.401e-02	6.695	2.58e-11 ***	popularity	7.600e-02	1.542e-02	4.929	8.74e-07 ***
genre	-2.783e-01	6.343e-02	-4.387	1.19e-05 ***	genre	-2.340e-01	7.785e-02	-3.006	0.00267 **
vote_count	2.661e-04	2.116e-05	12.578	< 2e-16 ***	vote_count	2.556e-04	2.583e-05	9.895	< 2e-16 ***
original_languageen	8.650e+00	6.720e-02	128.719	< 2e-16 ***	original_languageen	8.673e+00	7.887e-02	109.960	< 2e-16 ***
original_languagefr	8.114e+00	1.286e-01	63.094	< 2e-16 ***	original_languagefr	8.288e+00	1.559e-01	53.179	< 2e-16 ***
genre:vote_count	-1.033e-04	2.381e-05	-4.337	1.49e-05 ***	genre:vote_count	-7.376e-05	2.923e-05	-2.523	0.01169 *
--- Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					--- Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				
Residual standard error: 1.452 on 2909 degrees of freedom Multiple R-squared: 0.9788, Adjusted R-squared: 0.9788 F-statistic: 1.921e+04 on 7 and 2909 DF, p-value: < 2.2e-16					Residual standard error: 1.782 on 2909 degrees of freedom Multiple R-squared: 0.9684, Adjusted R-squared: 0.9683 F-statistic: 1.274e+04 on 7 and 2909 DF, p-value: < 2.2e-16				

Amount = 1

Amount = 2

Coefficients:					Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)		Estimate	Std. Error	t value	Pr(> t)
budget	3.212e-05	2.875e-06	11.174	< 2e-16 ***	budget	2.725e-05	5.256e-06	5.184	2.32e-07 ***
popularity	3.465e-02	1.695e-02	2.044	0.0411 *	popularity	3.115e-03	1.796e-02	0.173	0.862
genre	-6.125e-01	1.186e-01	-5.165	2.56e-07 ***	genre	-1.944e-01	2.178e-01	-0.892	0.372
vote_count	2.052e-04	2.695e-05	7.614	3.57e-14 ***	vote_count	3.083e-04	4.870e-05	6.331	2.81e-10 ***
original_languageen	8.951e+00	1.174e-01	76.244	< 2e-16 ***	original_languageen	8.977e+00	1.992e-01	45.059	< 2e-16 ***
original_languagefr	8.435e+00	2.708e-01	31.151	< 2e-16 ***	original_languagefr	9.190e+00	4.927e-01	18.654	< 2e-16 ***
--- Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					--- Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				
Residual standard error: 3.188 on 2910 degrees of freedom Multiple R-squared: 0.9045, Adjusted R-squared: 0.9043 F-statistic: 4593 on 6 and 2910 DF, p-value: < 2.2e-16					Residual standard error: 5.856 on 2910 degrees of freedom Multiple R-squared: 0.7448, Adjusted R-squared: 0.7443 F-statistic: 1416 on 6 and 2910 DF, p-value: < 2.2e-16				

Amount = 5

Amount = 10

Conclusions from Differential Privacy On Movies Data



Trade off between Privacy and Usability

- Differential privacy adds random noise to protect individual privacy this is controlled by parameter epsilon ϵ (controls level of privacy)
 - Smaller epsilon = more privacy, less usability
 - Greater epsilon = less privacy, more usability.
 - Aim for minimal impact on data accuracy.

In our case, adding a little bit of noise seemed to improve model, however it was not enough to fully protect privacy. However, the more noise added meaning the more information being protected, the less useful the model became.

Limitations

- Unable to figure out the appropriate balance between privacy and usability.
 - Consider data sensitivity and acceptable noise levels.
- Data set was not large enough for DP
 - Smaller dataset increase the challenge of maintaining both privacy and usability.
- Do not understand the relationship between noise and multicollinearity, therefore we are only stating our observations.

Future Research



- Change approach of adding noise to an exponential mechanism
 - Would involve transitioning from jitter method to better handle both numerical and categorical variables
- Understanding the ``diffpriv`` R package
 - Contains multiple methods to add noise: Laplace, Gaussian, Exponential, Bernstein.
 - Ideally implement this package to our Movies dataset to see how noise from these additional methods differ to jitter.
- Study and learn the underlying theory behind differential privacy
 - We had difficulty reading through the dense proof heavy papers but with time and gained knowledge we hope to be able to easier parse these papers.
- Research the impact of noise in multicollinearity

Thank You!